

# Realty360 Austin — Application Architecture Report

Version: 1.3 · Generated: July 7, 2026 · Market: Austin / Travis County, TX

This document describes what the application does today, how data flows through the system, key design decisions, and the planned evolution path. It reflects the codebase as of the feature/tax-data-pipeline branch (MLS SQLite cache, TCAD Neon Postgres cache, profile cache, multi-parcel TCAD handling, MLS+TCAD comp enrichment, rules-based flip engine).

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## 1. Executive Summary

Realty360 Austin is a full-stack property analysis tool for Travis County real estate investors and analysts. It combines:

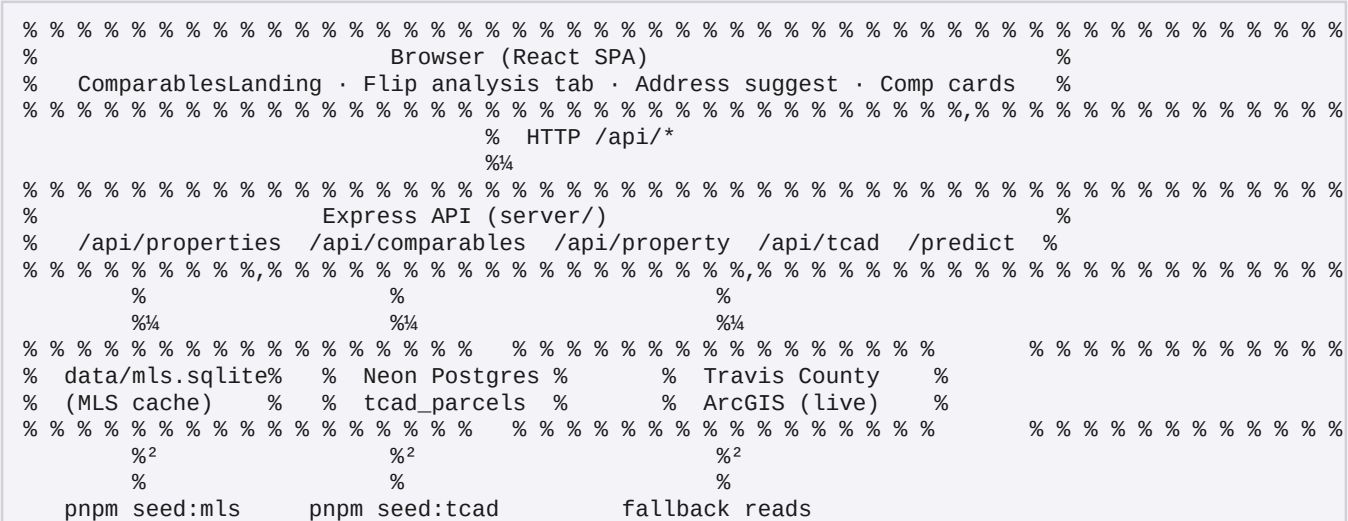
- MLS closed and open listings (sale comparables, subject lookup) - TCAD tax parcel data (assessed/appraised values, parcel centroids) - Unified comparables search (closed sales, active listings, optional tax-reference parcels; MLS comps enriched with TCAD tax fields) - Property profile composition with in-process cache and multi-parcel TCAD (taxCandidates) - Post-flip deal analysis (ARV from comps, rehab tiers, hold/financing costs with Policy B multi-parcel tax, margin viability)

The app is built as Vite + React (client) and Express (API), with shared TypeScript logic in shared/. Production serves a single Node process; development runs Vite on port 3000 proxying /api to Express on 3001.

What it is not (yet): a production MLS Grid proxy for the browser, a machine-learning ARV model, or a forced MLS!™TCAD crosswalk. Those are documented future phases.

## 2. System Architecture

### 2.1 High-Level Topology



### 2.2 Request Path (Development)

| Component           | Port | Role                                        |
|---------------------|------|---------------------------------------------|
| Vite dev server     | 3000 | UI, proxies `/api` to 3001                  |
| Express `dev:api`   | 3001 | JSON API, reads SQLite + Postgres           |
| `pnpm start` (prod) | 3000 | Bundled static + API (or API-only if no...) |

## 2.3 Shared Logic

Business rules live in shared/ so the same comp-matching, flip math, TCAD address scoring, and DTO composition run in scripts, API routes, and (where applicable) the client via @shared alias.

## 3. Repository Layout

| Path       | Purpose                                                         |
|------------|-----------------------------------------------------------------|
| `client/`  | React UI (Comparables landing, flip results, shadcn...          |
| `server/`  | Express app, route handlers                                     |
| `shared/`  | Domain logic: MLS, TCAD, comparables, property-profile, flip    |
| `scripts/` | Seed jobs (`seed-mls`, `seed-mls-open`, `seed-tcad`), `smoke... |
| `data/`    | Gitignored runtime data (`mls.sqlite`, raw JSON dumps)          |
| `docs/`    | Roadmap and architecture documentation                          |

## 4. Data Sources & Storage

### 4.1 MLS (Multiple Listing Service)

| Aspect             | Detail                                                            |
|--------------------|-------------------------------------------------------------------|
| Upstream           | MLS Grid API (`https://api-demo.mlsgrid.com/v2` or production...) |
| Originating system | `actris` (Austin Board of Realtors) — `shared/mls/constants.ts`   |
| Local store        | SQLite `data/mls.sqlite`, table `properties`                      |
| Seed commands      | `pnpm seed:mls` (closed residential), `pnpm seed:mls:open`...     |
| Runtime            | API reads SQLite only; no live MLS Grid on user requests          |
| Token              | `MLS_GRID_TOKEN` or `VITE_MLS_GRID_TOKEN` in...                   |

Row shape: normalized address, lat/lon, beds/baths/sqft, list/close price, close date, pool, garage, lot acres, property condition, status.

### 4.2 TCAD (Travis Central Appraisal District)

| Aspect       | Detail                                                        |
|--------------|---------------------------------------------------------------|
| Upstream     | ArcGIS MapServer layer `TCAD/MapServer/0` (full county...)    |
| Local store  | Postgres `tcad_parcel` (Neon) via `DATABASE_URL`              |
| Seed command | `pnpm seed:tcad` (~387k parcels, paginated ETL)               |
| Runtime      | Cache-first in `shared/tcad/client.ts`; live ArcGIS fallback  |
| Coverage     | Travis County only — Williamson/Georgetown addresses retur... |

Fields cached: PROP\_ID, geo\_id, situs address, tax values (appraised, market, assessed), acres, WGS-84 centroid.

### 4.3 Data Independence Principle

MLS and TCAD are intentionally separate. There is no required listing\_key !" prop\_id crosswalk in v1. Profiles compose both sources at request time when possible; partial profiles are valid.

## 5. Core User Flows

### 5.1 Comparables Search

1. User enters address (with typeahead from MLS SQLite). 2. UI calls GET /api/comparables/unified with radius, recency, optional includeTcad, optional propId. 3. Server loads subject via fetchPropertyProfileCached (MLS + TCAD profile-first); resolves MLS row for comp filters. 4. Returns sections: - Closed sales (MLS) — sale comps, each enriched with TCAD taxValue and tcadAcres when a parcel matches - Active listings (MLS open status) — same TCAD enrichment on listing comps - Tax references (TCAD radius, includeTcad=true) — reference only, not sale comps 5. Response includes subjectProfile for subject card tax display and flip cache reuse.

Subject card: shows TCAD tax value, acres, deed date when single match; lists all taxCandidates with combined tax when ambiguous; optional parcel picker re-runs search with propId.

TCAD enrichment (closed + open comps): one Postgres radius prefetch per search

(findTcadParcelsInRadiusBBox), then in-memory address scoring and street-aware nearest-parcel fallback.

Coords for matching prefer MLS lat/lon; TCAD centroid fills gaps. UI citation: MLS + TCAD when enriched.

Comps are bucketed into exclusive mile rings (0–0.5, 0.5–1, 1–2 mi, ...) with match % from beds/baths/distance.

### 5.2 Property Profile

1. GET /api/property/profile?address= or ?propId= (or both) 2. fetchPropertyProfile() / fetchPropertyProfileCached() loads MLS by address and TCAD by address or propId. 3. composePropertyProfile() merges identifiers, physical, MLS sale, tax, location. 4. TCAD is always attempted for tax data even when MLS supplies coordinates. 5. Single TCAD match !' tax set, tcadMatch.status: "single". 6. Ambiguous TCAD match !' tax: null, taxCandidates[], tcadMatch.status: "ambiguous" (e.g. duplex — multiple prop\_id at same situs). 7. No TCAD match !' MLS-only profile, tcadMatch.status: "not\_found"; comparables and flip continue without tax fields. 8. Address matching tolerates MLS-style queries (504 Bramble Dr, Austin, TX 78745) by excluding city/state tokens from situs scoring (shared/tcad/address-query.ts).

Profile cache: in-process Map, 30 min TTL, keys address:{norm} and/or propId:{id}address:{norm}. Comparables search warms cache; flip reuses same profile without re-fetching TCAD.

### 5.3 Flip Analysis

1. User runs comparables search, switches to Flip analysis tab. 2. Enters purchase price and rehab scope tier (cosmetic / moderate / full). 3. POST /api/predict/flip in enriched mode: - Loads property profile via fetchPropertyProfileCached (address only — preserves taxCandidates for Policy B even if UI selected a propId for display) - Derives ARV from similar closed MLS comps (median \$/sqft × subject sqft) - Computes rehab, closing, hold, financing - Hold tax basis (Policy B): single parcel !' assessed !' appraised; ambiguous !' sum assessed across all taxCandidates; no TCAD !' purchase price - Returns viability band: strong / marginal / weak / negative

Manual mode: user supplies explicit arv + livingAreaSqft without subject lookup.

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## 6. API Surface

| Method | Endpoint                                  | Purpose                                  |
|--------|-------------------------------------------|------------------------------------------|
| GET    | <code>`/api/health`</code>                | Liveness check                           |
| GET    | <code>`/api/properties/suggest`</code>    | Address typeahead (MLS)                  |
| GET    | <code>`/api/properties/by-address`</code> | Single MLS property                      |
| GET    | <code>`/api/properties/by-radius`</code>  | Closed comps in radius                   |
| GET    | <code>`/api/comparables/unified`</code>   | Closed + open + optional TCAD; option... |
| GET    | <code>`/api/property/profile`</code>      | Composed MLS + TCAD profile              |
| GET    | <code>`/api/tcad/property`</code>         | TCAD parcel by propld or address         |
| POST   | <code>`/api/predict/flip`</code>          | Rules-based flip prediction              |

All routes are mounted in `server/app.ts`.

## 7. Domain Modules

### 7.1 `shared/mls/`

- RESO property types, MLS Grid client, address normalization - `scripts/mls-db.ts`: SQLite schema, radius queries, address resolution

### 7.2 `shared/tcad/`

- ArcGIS client with retry/timeout - Address fuzzy matching (`address-query.ts`) - Postgres cache (`db.ts`) - Parcel centroids from polygon geometry

### 7.3 `shared/comparables/`

- `match-score.ts` — mile rings, match %, distance score - `comp-record.ts` — unified `CompRecordDto` with source, `compRole`, `taxValue`, `tcadAcres` - `tcad-enrichment.ts` — batch TCAD prefetch + in-memory match for MLS sale/listing comps - `unified-search.ts` — orchestrates MLS + TCAD comp sections and enrichment

### 7.4 `shared/property-profile/`

- `fetch-profile.ts` — enrichment orchestration (MLS + TCAD, ambiguous handling) - `compose.ts` — merges partial MLS/TCAD into `PropertyProfileDto` - `profile-cache.ts` — in-process TTL cache keyed by address and/or `propld` - `tcad-tax.ts` — `pickProfileTaxValue`, `taxHoldBasisFromProfile` (Policy B sum for multi-parcel) - Provenance and completeness scoring

### 7.5 `shared/flip/`

- `config.ts` — Travis County rehab tiers, closing %, hold/financing defaults - `arv-comps.ts` — similarity filter, tier-based \$/sqft slice (top half / quartile) - `deal-costs.ts` — deterministic cost stack - `predict-flip.ts` — end-to-end prediction pipeline

## 8. Flip Prediction Engine

### 8.1 ARV (After Repair Value)

Source: MLS closed sales within radius, filtered by:

- Recency (`maxAgeMonths` on `close_date`) - Similarity:  $\pm 25\%$  sqft, beds/baths within delta rules - Rehab tier slice: cosmetic uses all similar; moderate uses top 50% \$/sqft; full uses top 25%

```
ARV = median(pricePerSqft of slice) × subjectLivingAreaSqft
```

Override: request body `arv` != manual mode.

### 8.2 Cost Stack

| Component                          | Basis                                                     |
|------------------------------------|-----------------------------------------------------------|
| Rehab                              | tier \$/sqft × living area (sqft from MLS profile)        |
| Buy closing                        | 2.5% of purchase price (user input)                       |
| Hold — property tax                | Policy B: sum assessed across `taxCandidates` when...     |
| Hold — insurance / utilities / HOA | fixed monthly amounts from deal config                    |
| Financing                          | hard-money style: LTC ratio, points, interest during hold |
| Sell closing                       | 7% of ARV (ARV from MLS comps)                            |

### 8.3 Viability Bands

| Net margin % (of ARV) | Label    |
|-----------------------|----------|
| "e 12%                | strong   |
| 6–12%                 | marginal |
| 0–6%                  | weak     |
| < 0%                  | negative |

### 8.4 Explicit Non-Goals

- TCAD appraised value is not used as ARV or sale comp - Tax values affect hold cost only (when TCAD resolves)

## 9. Key Design Decisions

| Decision                              | Rationale                                                        |
|---------------------------------------|------------------------------------------------------------------|
| MLS in SQLite, not live API           | Fast radius queries; avoids MLS Grid rat...                      |
| TCAD in Postgres (Neon)               | ~373k parcels; eliminates per-request...                         |
| Cache-first TCAD with ArcGIS fallback | Works before seed completes; resilient t... tcad-arcgis` on DTOs |
| MLS comp TCAD enrichment              | Sale/listing comps show unified tax field...                     |
| Profile cache (30 min TTL)            | Comparables warms profile; flip reuses...                        |
| Multi-parcel Policy B                 | Duplex/multi-`prop_id` situs: hold tax...                        |
| MLS-only fallback                     | TCAD `not_found` != valid MLS profile; ho...                     |
| MLS-style address scoring             | City/state stripped from TCAD situs...                           |
| Batch comp enrichment                 | Single radius prefetch + in-memory inde...                       |
| No forced MLS!TCAD link in v1         | Address matching is messy; partial...                            |
| TCAD tax comps are reference only     | Appraised "" market sale price; prevents...                      |
| Rules engine before ML                | Auditable math; same API contract for...                         |
| Shared TypeScript domain layer        | One source of truth for comps, flip, and...                      |
| Enriched vs manual flip modes         | Supports full subject lookup or...                               |
| Dedupe PROP_ID on TCAD seed           | ArcGIS pages can repeat prop_ids...                              |

## 10. Operational Commands

```

pnpm install
pnpm seed:mls          # MLS closed !' data/mls.sqlite
pnpm seed:mls:open     # Active/pending into same DB
pnpm seed:tcad         # Full county !' Neon tcad_parcel
pnpm dev:api           # API :3001
pnpm dev               # UI :3000
pnpm smoke:flip        # API/engine regression checks
pnpm docs:architecture-pdf # Regenerate docs/ARCHITECTURE_REPORT.pdf
pnpm build && pnpm start # Production-style single server

```

Environment (.env.local):

- VITE\_MLS\_GRID\_TOKEN / MLS\_GRID\_TOKEN — MLS Grid bearer token - DATABASE\_URL — Neon Postgres for TCAD cache (tcad\_parcel). Must match the Neon project/branch you intend (verify ep-... hostname in connection string vs dashboard).

## 11. Roadmap & Gaps

### Completed (Phases 1–4 + TCAD platform)

- MLS schema extensions, mile-bucket comps, unified comparables API - Property profile composition, TCAD propld/address APIs, profile cache, multi-parcel taxCandidates - Subject card TCAD tax display; optional propld parcel picker - Flip rules engine v0, flip UI on comparables landing (FlipPredictionResults) - Policy B hold tax (taxHoldBasisFromProfile — sum assessed for ambiguous parcels) - TCAD Neon ETL (pnpm seed:tcad, ~373k parcels), cache-first reads (shared/tcad/db.ts) - MLS sale/listing comp enrichment with TCAD tax value + acres (tcad-enrichment.ts) - MLS-style TCAD address matching (city/state excluded from scoring) - Flip smoke tests (pnpm smoke:flip)

### MLS !" TCAD coverage (July 2026 audit)

Of 6,139 distinct MLS addresses crosswalked against tcad\_parcel:

| Outcome                  | Count |
|--------------------------|-------|
| Single TCAD parcel       | 1,573 |
| No TCAD match (MLS-only) | 4,539 |
| Ambiguous (multi-parcel) | 27    |

Example multi-parcel test: 1613 W Braker Ln #B, Austin, TX 78758 (2 TCAD parcels, same situs). Example MLS-only: 507 Hammack Dr (MLS present; TCAD situs numbering gap 505!600 on Hammack).

### Planned

| Item                           | Description                                                        |
|--------------------------------|--------------------------------------------------------------------|
| MLS !" TCAD crosswalk          | Precomputed `listing_key` !" `prop_id` links for faster comp joins |
| Deal ledger                    | Labeled flips for ML training                                      |
| Python ML ARV model            | Same `POST /api/predict/flip` contract                             |
| `geold` lookup                 | Alternate TCAD key                                                 |
| Coordinate-based TCAD fallback | When address `not_found` but MLS lat/lon exists, match neare...    |
| Monthly TCAD refresh           | Cron re-run `seed:tcad`                                            |

### Known Limitations

- MLS seed cap (~2k–5k rows configurable) — not full ACTRIS history - Non-Travis addresses have MLS data but no TCAD tax - ~74% of MLS addresses have no TCAD situs match in current cache (address gaps, situs numbering mismatches) - Multi-parcel ambiguous is rare (~27 MLS addresses); most duplexes either score to one winner or have no TCAD situs - Condos/townhomes may lack unit-level TCAD situs — nearest same-street parcel used for comp enrichment - Flip ARV requires sufficient similar closed comps in radius - Home page

calculator may still use embedded constants if MLS token unset

## 12. Appendix: Diagrams

### 12.1 Property Profile Composition

```

% % % % % % % % % % % % % % % % %
%   User query   %
%   address/propId %
% % % % % % % % % % % % % % % % %
%
% % % % % % % % % % % % % % 4% % % % % % % % % % % % %
% % % % % % % % % % % % % % 4%
% % % % % % % % % % % % % % % % % % % % % % % % % % % %
%   MLS SQLite   %           %   TCAD cache   %
%   resolve addr %           %   or ArcGIS    %
% % % % % % % % % % % % % % % % % % % % % % % % % % % %
%
% % % % % % % % % % % % % % % % % % % % % % % % % % % %
% % % % % % % % % % % % % % % % % % % % % % % % % % % %
%   composeProperty %
%   ProfileDto      %
% % % % % % % % % % % % % % % % %
%
```

### 12.2 Flip Prediction Pipeline

```

POST /api/predict/flip
%
% % enriched? % % %° fetchPropertyProfileCached (MLS + TCAD)
%
% % resolve livingAreaSqft (profile or override)
%
% % ARV: manual override OR comp median $/sqft × sqft
% % % % MLS closed comps in radius (SQLite)
%
% % taxBasis (Policy B):
% % % % single tax !' assessed !' appraised
% % % % taxCandidates !' SUM assessed (no purchase fallback)
% % % % not_found !' purchasePrice
%
% % costs: rehab + closing + hold + financing
%
% % margins + viability % % %° FlipPredictionResponse
```

### 12.3 TCAD Read Path

```

API handler (profile / tcad / unified / flip)
%
% %
shared/tcad/client.ts
%
% % DATABASE_URL set?
% % % % YES % % %° shared/tcad/db.ts (Postgres)
% % % % propId / address candidates / radius
% % % % miss or error % % %° live ArcGIS query
%
% % NO % % %° live ArcGIS only
```

### 12.4 Comp Roles (Unified API)

